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Numerical Simulations of the Ising model through Monte Carlo Markov Chains

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Abstract

This work presents the numerical study of the ferromagnetic Ising model in one, two, and three dimensions to observe phase transitions as a function of the lattice size. The system configurations were obtained by sampling a Markov Chain at fixed temperature generated with the Metropolis-Hastings algorithm following an analysis of its thermalisation and autocorrelation time. The critical temperature was estimated in the two and three dimensional cases by analyzing the finite size scaling of the magnetization. Analogously, the absence of a finite temperature phase transition was found in the one dimensional case, in accordance with analytical results.

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A brief history of the Ising model

Does this chapter really need to exist?

Chapter 1

Theoretical background

1.1 General overview

The Ising model is a mathematical model of a ferromagnet. [TODO: add context].

The fundamental assumption of the model is that the solid can be represented with every ion fixed in space and only interacting with its closest neighbours (and, optionally, an external magnetic field). Such system is described by the classical Hamiltonian

$$H = -\sum_{\langle i,j \rangle} J_{ij}\sigma_i\sigma_j - \mu\sum_i h_i\sigma_i \tag{1.1}$$

where:

- σ_i is the spin value of the ion of the i-th lattice site.
- J_{ij} is a coupling factor between the i-th and j-th lattice sites.
- The sum over $\langle i, j \rangle$ is meant to be over closest neighbours.
- h_i is the magnetic field in the i-th site.
- μ is the magnetic moment of the ions.

This is the special case of the more general quantum Hamiltonian

$$\hat{H} = -\sum_{\langle i,j \rangle} J_{ij}\hat{\sigma_i^z}\hat{\sigma_j^z} - \mu\sum_i \left(h_{i,x}\hat{\sigma_i^x} + h_{i,y}\hat{\sigma_i^y} + h_{i,z}\hat{\sigma_i^z}\right) \tag{1.2}$$

where σ_i are replaced by the Pauli matrices of the i-th site.

In this text only $h_{i,z}$ will ever be assumed $\neq 0$, which makes every operator commute with each other in the canonical basis, which means equations 1.1 and 1.2 are the same.

Given that the quantities $N =$ number of particles, $V = Na =$ volume of the lattice (where a is the unimportant lattice constant that can be set to 1) are fixed, if the system is in thermal equilibrium with a reservoir at temperature T then the set of microstates of the system is described by the canonical ensemble. In this case, the probability of the system to be in a microstate σ is given by the Boltzmann distribution

$$P_\beta(\sigma) = \frac{e^{-\beta H(\sigma)}}{Z_\beta} \tag{1.3}$$

where Z_β is the canonical partition function

$$Z = \sum_i e^{-\beta E_i} = \sum_{\sigma_i} e^{-\beta \sum_{\langle i,j \rangle} J_{ij}\sigma_i\sigma_j - \beta\mu \sum_i h_i\sigma_i} \tag{1.4}$$

All the same, given an operator $\hat{O}$ its mean is intended to be

$$\left\langle \hat{O} \right\rangle_\beta = \sum_\sigma \hat{O}\left(\sigma\right) P_\beta\left(\sigma\right) \tag{1.5}$$

which is the canonical average.

The minus in front of the first sum in equations 1.1 and 1.2 are conventional. Varying the sign of the coefficients J_{ij} two differently behaving systems are described:

1. $J_{ij} > 0$: ferromagnetic
2. $J_{ij} < 0$: antiferromagnetic
3. $J_{ij} = 0$: non-interacting

Historically, the Ising model is conventionally intended to be ferromagnetic. For simplicity sake, J_{ij} along the constants k_B, $\hbar$, h_i, μ will be set $= 1\ \forall i, j$, when not null.

1.2 Phase transitions

It's been stated that the system described by the Ising model is represented by the canonical ensemble, which is governed by the Helmoltz free energy. This, like every other thermodynamic potential, can be expressed as a function of the partition function $Z = Z(T, h)$:

$$F = E - TS = -\frac{\ln\left(Z\right)}{N\beta} \tag{1.6}$$

Formally, a phase transition is defined as a non-analyticity in the free energy in the thermodynamic limit. To observe such a non-analyticity one can study the partial derivative

$$M = -\frac{\partial F}{\partial H} \tag{1.7}$$

that takes the name of magnetization.

This is a handy observable in the numerical approach beacuse its average is easily obtainable by averaging the spin values of the lattice sites:

$$\langle M \rangle = \frac{1}{N} \sum_i \sigma_i \tag{1.8}$$

and studying its behaviour as a function of the temperature T and the external magnetic field h one can identify the phase transitions.

It should be noted that this definitions formally hold even in the case of no external magnetic field. This is true only in the thermodynamic limit, where the system is infinitely large and the magnetization can be non-zero even when h vanishes.

In a finite system the Hamiltonian is symmetric under the transformation $\sigma_i \to -\sigma_i$, and this causes the magnetization to be zero when $h = 0$ because

$$\langle M \rangle = \left. \frac{\partial \frac{\ln(Z)}{\beta N}}{\partial h} \right|_{h=0} = 0 \ \forall T$$

The solution to this problem is

[TODO: i fear this can be greatly expanded on]

1.3 Solutions

1.3.1 One dimensional case

The one dimensional Ising model is exactly solvable with and without an external magnetic field. The magnetization is given by:

[TODO: discuss solutions]

1.3.2 Two dimensional case

1.3.3 Three dimensional case

Chapter 2

Methods

2.1 Data representation

[TODO: discuss floating point precision, data structures, etc.]

2.2 Monte Carlo Markov Chains

2.2.1 Generation

[TODO: discuss Metropolis-Hastings algorithm]

2.2.2 Analysis

After having generated the Markov Chain, the one of the analysis is a problem composed of two parts: the thermalisation and the autocorrelation time. The first one is the number of steps that the Markov Chain needs to reach a stationary distribution, while the second one is the number of steps that need to exist between successive samples before they can be considered independent.

[TODO: discussion of the autocorrelation function and its properties. An exponential fit after a certainb number of steps is only fitting noise.]

NOTA: Figure 2.1 prove the need of selecting a proper range to fit over. This is given by the linear portion of the following plot.

[TODO: Didnt manage to make use of this]

[TODO: discussion of Sokal's method]

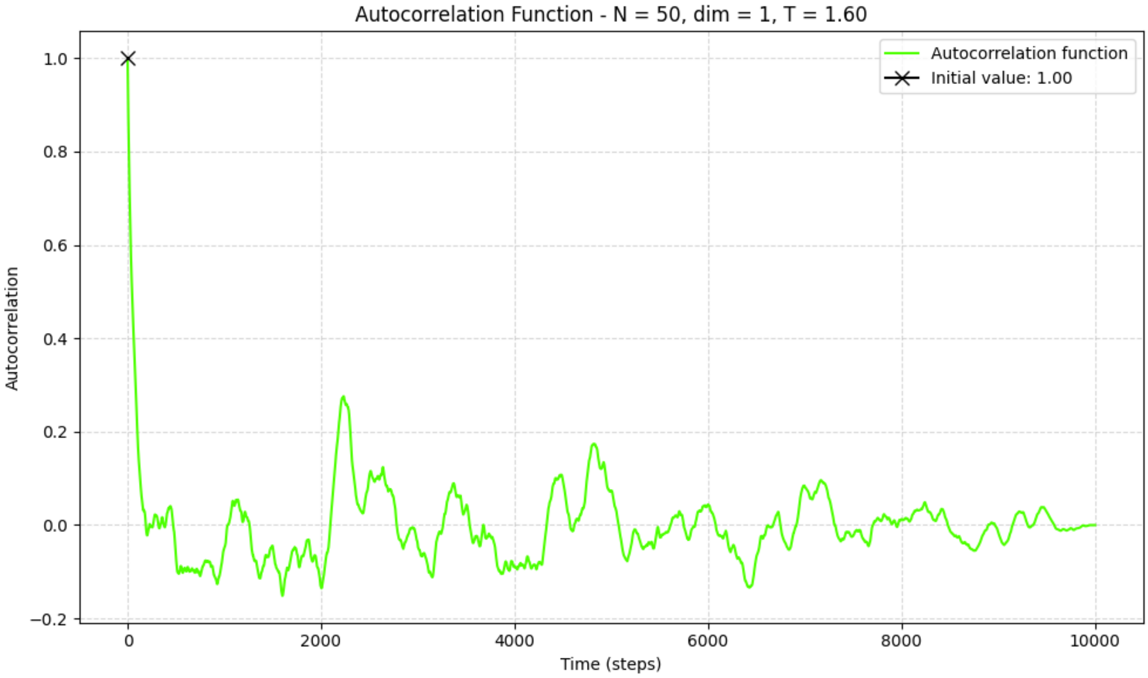

Figure 2.1: Example of autocorrelation function for a Markov Chain.

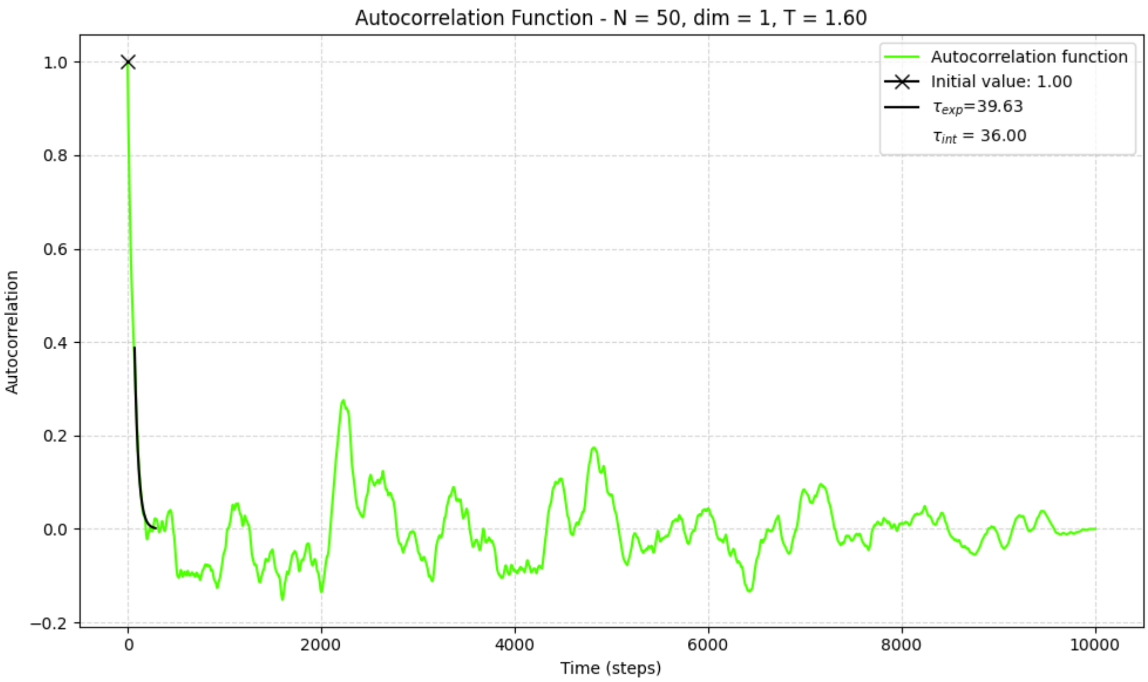

Figure 2.2: Fitting to find τ_{exp}.

[TODO: discussion of the data blocking method]

Chapter 3

Results

3.1 1D Ising model

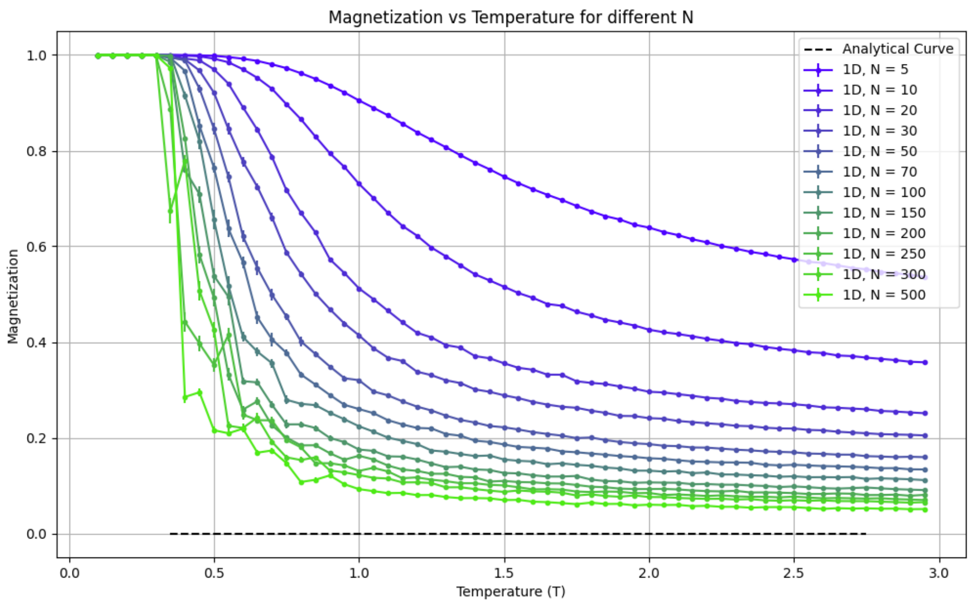

Figure 3.1: Convergence of magnetization for the 1D Ising model.

3.2 2D Ising model

3.3 3D Ising model

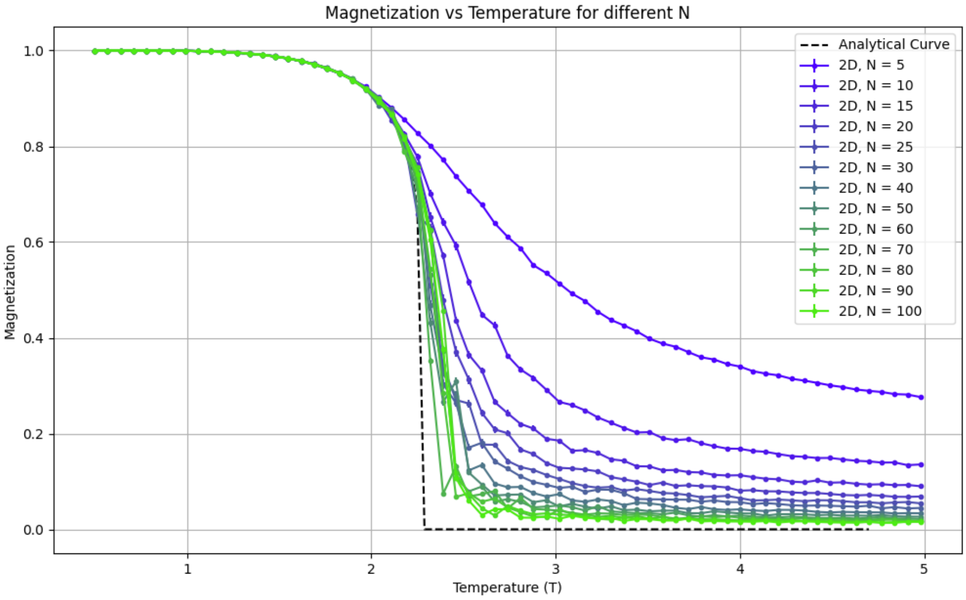

Figure 3.2: Convergence of magnetization for the 2D Ising model.

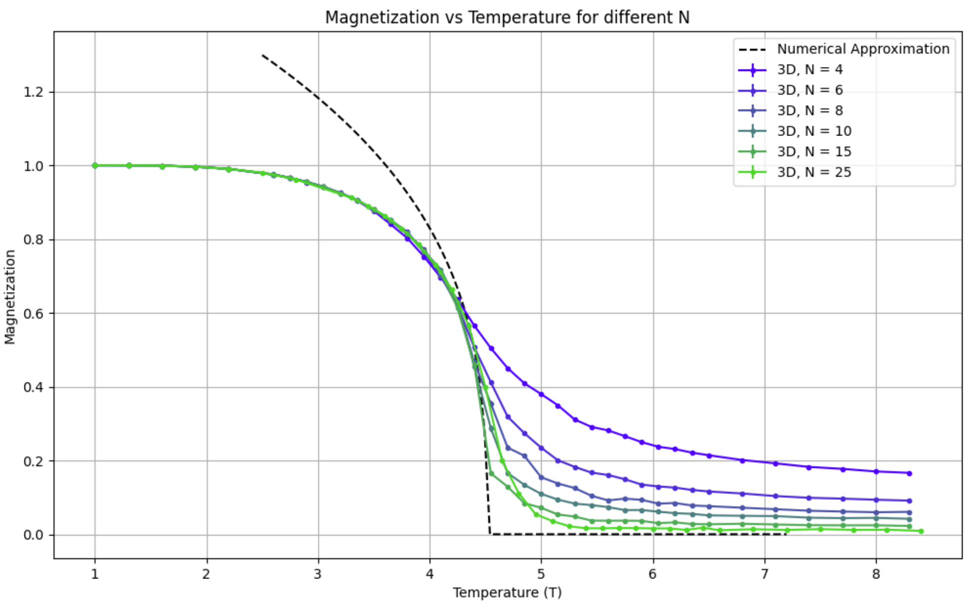

Figure 3.3: Convergence of magnetization for the 3D Ising model.

Appendix A

Appendix

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Bibliography

[1] Author, Title, Journal/Book, Year.